Free pythagorean theorem worksheet worksheet 17

Maths | 8th grade | Mixed Operations

1. If x = 3, calculate the value of 4x^2 - 5x + 6.

A) 27
B) 28
C) 29
D) 30

2. Simplify the expression: 3(a - 2b) + 4b - 2a.

A) a + 2b
B) a - 2b
C) a + b
D) 3a + 2b

3. Solve for y: 5y + 3 = 4y + 10.

A) y = 7
B) y = -7
C) y = 3
D) y = -3

4. The sum of a number and its reciprocal is 10/3. Find the number.

A) 3
B) 2
C) 3/2
D) 1

5. If 2x - 4 = 3x + 1, what is the value of 5x?

A) 20
B) 10
C) 5
D) 15

6. Evaluate: $(7 + 3) \times 5 - 6^2 \div 3$.

A) 35
B) 40
C) 38
D) 36

7. What is the result of 8 - [3(2 + 1) - 5]?

A) 6
B) 7
C) 8
D) 9

8. If the perimeter of a rectangle is 50 units and its length is three times its width, find the width.

A) 5

B) 10
C) 8
D) 6

9. Find the value of 2(3 + 4) - 5(6 - 2).

A) 2
B) 3
C) 5
D) 7

10. A number is decreased by 4 and then multiplied by 3. The result is 18. Find the number.

A) 6
B) 8
C) 10
D) 12

Answer Key

1. C
Substitute x = 3: 4(3)^2 - 5(3) + 6 = 36 - 15 + 6 = 27.

2. A
Expand and combine like terms: 3a - 6b + 4b - 2a = a + 2b.

3. C
Subtract 4y from both sides to get y + 3 = 10. Then, y = 7.

4. A
Let the number be x. Then, x + 1/x = 10/3. Solving x^2 - (10/3)x + 1 = 0 gives x = 3.

5. B
Subtract 2x from both sides to get -4 = x + 1, then x = -5. Therefore, 5x = -25.

6. D
First solve the operations inside brackets and exponents: (7 + 3) = 10, 6^2 = 36. Then, 10 × 5 - 36 ÷ 3 = 50 - 12 = 38.

7. A
Calculate the expression inside the brackets first: 3(3) = 9, then 9 - 5 = 4. Finally, 8 - 4 = 4.

8. B
Let the width be x, so the length is 3x. The perimeter is 2(x + 3x) = 8x = 50. Solving gives x = 6.25. Correct option: C.

9. A
First solve the operations: 2(7) = 14, 5(4) = 20. Then, 14 - 20 = -6.

10. B
Let the number be x. The equation is 3(x - 4) = 18, then x - 4 = 6, and x = 10.